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Case 11-2025: A 79-Year-Old Woman with Cough and Weight Loss

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PRESENTATION OF CASE

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N Engl J Med 2025;392:1532-43.

DOI: 10.1056/NEJMcpc2412520

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CME



Dr. Shambhavi Singh (Medicine): A 79-year-old woman was evaluated in the primary care clinic of this hospital because of cough and weight loss.

The patient had been in her usual state of health until 4 weeks before the current presentation, when dry cough, coryza, sore throat, and fatigue developed. Coryza and sore throat resolved, but cough and fatigue persisted. She was evaluated by her primary care physician.

In the primary care clinic, the patient reported ongoing cough that had become productive of sputum and had begun interfering with sleep; she attributed the ongoing fatigue to disrupted sleep. No dyspnea, pleurisy, or hemoptysis was present. The weight had decreased by 10.4 kg during the past year. There were no night sweats. She had traveled to South Africa 5 years before the current presentation, to South America 4 years before the current presentation, and to Southeast Asia 6 months before the current presentation. During the trip to Southeast Asia, urinary frequency and dysuria developed. The patient was treated with an unknown antibiotic agent, and her symptoms resolved. Persistent nonbloody diarrhea developed after treatment with the antibiotic; bowel movements were more frequent and loose after she consumed certain foods (coffee and ice cream). After adjustments in diet, diarrhea abated but did not resolve.

The patient had a history of osteoporosis, recurrent urinary tract infections, and primary hyperparathyroidism, for which she had undergone parathyroidectomy. Medications included calcium carbonate, cholecalciferol, and a multivitamin. There were no known adverse reactions to medications. She was a retired professional, was widowed, and lived alone in a suburb of Boston; she helped provide care for her mother. She traveled frequently to California to visit her family. She was a lifelong nonsmoker and drank alcohol occasionally. Her mother had had lymphoma, which had been diagnosed 25 years earlier; the patient did not know the type of lymphoma or the specific details of treatment that had ultimately led to remission. Her mother had a relapse 8 years earlier, and after chemotherapy and radiation therapy, she was again in remission.

On examination, the temporal temperature was 35.6°C, the blood pressure 116/69 mm Hg, the pulse 81 beats per minute, the respiratory rate 16 breaths per minute, and the oxygen saturation 98% while the patient was breathing ambient air. She appeared thin. The breath sounds were diminished at the right lung base, and dullness was noted on percussion. There was no palpable cervical, supraclavicular, or axillary lymphadenopathy. The abdomen was nondistended and soft, with no tenderness on palpation; she had no evidence of splenomegaly. There was no clubbing.

The blood levels of electrolytes, glucose, alanine aminotransferase, aspartate aminotransferase, alkaline phosphatase, albumin, and protein were normal, as were the complete blood count with differential count and the results of tests of kidney function. The blood level of lactate dehydrogenase was 227 U per liter (reference range, 110 to 210), and the blood level of total protein was 8.0 g per deciliter (reference range, 6.4 to 8.3). The erythrocyte sedimentation rate was 55 mm per hour (reference range, 0 to 29), and the level of C-reactive protein was 11.9 mg per liter (reference value, >8.0). Imaging studies were obtained to help determine the cause of the patient's cough and to further assess the abnormal findings on pulmonary examination.

Dr. Ryan Willett: Two days later, chest radiography (Fig. 1) revealed a moderate-sized right pleural effusion and airspace opacities in the right middle and lower lobes. The right cardio-mediastinal silhouette was partially obscured.

Dr. Singh: Six days later, thoracentesis was performed, and 1.3 liters of pleural fluid were removed. The fluid was yellow and moderately turbid. There were 14,000 red cells per microliter and 6342 nucleated cells per microliter; 10% of the nucleated cells were neutrophils, 30% were lymphocytes, 1% were monocytes, 5% were macrophages, and 54% were unclassified cells. The pleural fluid level of total protein was 5.6 g per deciliter, and the pleural fluid level of lactate dehydrogenase was 230 U per liter.

Additional diagnostic tests were performed.

DIFFERENTIAL DIAGNOSIS

Dr. Leigh H. Simmons: I participated in the care of this patient and am aware of the final diagnosis.

At an annual wellness visit, this 79-year-old woman reported 4 weeks of cough and fatigue, as well as weight loss during the past year. Examination findings were consistent with a unilateral pleural effusion, which was confirmed on

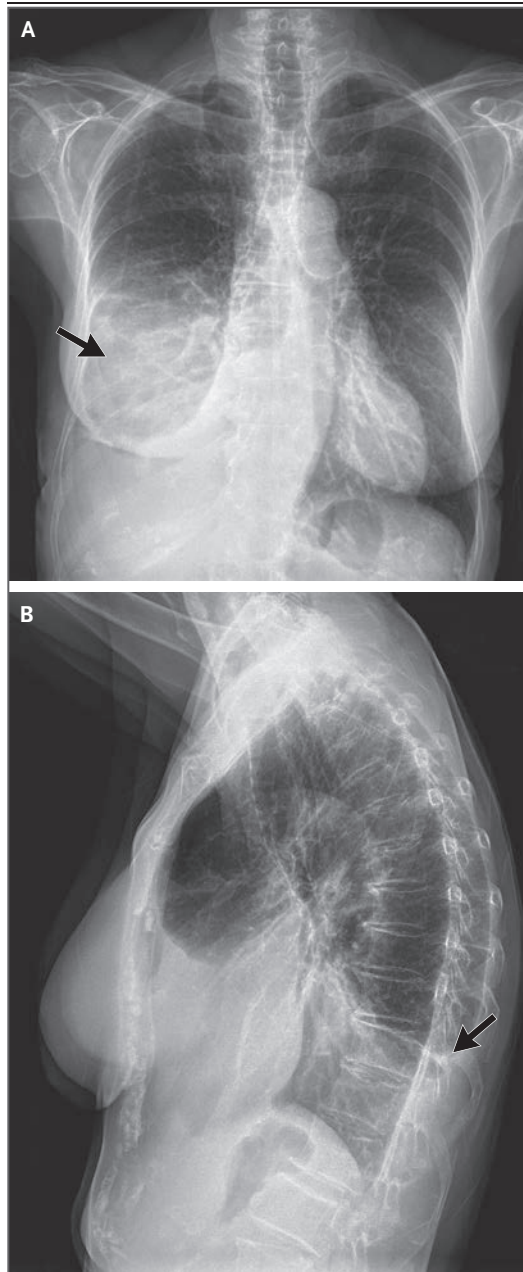


Figure 1. Chest Radiographs.

Chest radiographs show a moderate-sized right pleural effusion and airspace opacities in the right middle and lower lobes (Panels A and B, arrows). The right cardio-mediastinal silhouette is partially obscured.

chest radiography. The patient had returned to the clinic later that week to discuss her ongoing cough and review the results of chest radiography. When I had evaluated this patient, I was concerned about the new pleural effusion, and I explained to her that our next steps in the evaluation of her illness would focus on the cause of the fluid accumulation.

PLEURAL EFFUSION

When I was in medical school, I often heard the adage “Don’t let the sun set on an untapped pleural effusion” from my supervisors, and it was expected that the house officers would perform diagnostic thoracentesis as soon as a pleural effusion was discovered. The reason for this sense of urgency was the risk of complications and death associated with an untreated parapneumonic effusion.¹ However, things have changed substantially over the past two decades. The use of point-of-care ultrasonography has improved our ability to diagnose and characterize effusions, and at this hospital, we collaborate with interventional radiologists and pulmonologists to perform thoracentesis with imaging guidance, usually on a scheduled basis. However, the discovery of a new pleural effusion is almost always an indication for fluid sampling — even if not performed urgently — in order to make a diagnosis expediently.

In the case of this patient, diagnostic thoracentesis was performed soon after her visit with me to help guide the next steps in her evaluation. Pleural fluid analysis showed that the fluid met all three of Light’s criteria for exudative effusion.² In addition, the fluid was highly cellular, with unclassified cells accounting for 54% of the total nucleated cells. Once I knew the characteristics of the fluid, I focused my differential diagnosis on the three main causes of exudative pleural effusion: infection, inflammation, and cancer.

INFECTION

I considered the possibility that the patient had a parapneumonic effusion, given her history of cough, coryza, sore throat, and fatigue, as well as the airspace opacities seen in the right lung on chest radiography. However, she did not appear to be as ill as I would expect if she had a parapneumonic effusion resulting from bacterial pneumonia. Further radiographic evaluation of

the lung parenchyma after thoracentesis would be helpful in evaluating for underlying pneumonia. The patient’s travel history to Southeast Asia prompted consideration of tuberculosis, and her weight loss and diarrhea were consistent with this diagnosis. Although no opacities were seen in the upper lobes on chest radiography, tuberculosis could not be ruled out on the basis of this result.

NONINFECTIOUS CAUSES OF EXUDATIVE PLEURAL EFFUSION

Pulmonary embolism is a common clinical condition, and a substantial portion of patients with pulmonary emboli have an associated pleural effusion. However, this patient’s systemic symptoms did not fit well with this diagnosis, she did not have respiratory compromise, and the pleural effusion was larger than I would expect if it was due to pulmonary embolism, which typically causes small effusions.

I also considered the possibility that the patient had a systemic inflammatory illness that had caused the pleural effusion. Given her weight loss and fatigue, systemic lupus erythematosus was a consideration, but she had no arthritis, anemia, or renal disease that would support this diagnosis.

Drug-induced pleural effusion should be considered in patients with exudative pleural effusion. More than 30 drugs have been reported to cause pleural disease; therefore, taking a careful medication history is important, given that even intermittent use of these medications can cause disease.³ Some commonly prescribed medications associated with pleural disease include nitrofurantoin, minoxidil, amiodarone, and hydralazine. Although this patient did not take many medicines, she noted that she had recurrent urinary tract infections and had received treatment with antibiotics during the 6 months before presentation. Although she did not recall the name of the antibiotic she took while traveling in Southeast Asia, it is possible that she was treated with nitrofurantoin, a medicine that many patients take as needed for the treatment or prevention of urinary tract infections.

CANCER

Finally, I considered the possibility that cancer was causing the pleural effusion, and I ultimately determined that this was the most likely

explanation for the patient's presentation. Her weight loss and fatigue were consistent with cancer. A primary lung cancer was a top consideration, and although no obvious lung parenchymal lesions were noted on chest radiography, chest computed tomography (CT) was needed to evaluate for lung or pleural tumors. Lymphoma was a consideration because it can manifest with primary pleural effusion. I also considered the possibility of metastatic disease related to breast or ovarian cancer as a cause of her effusion. The patient had chosen to continue breast cancer screening with mammograms after 75 years of age; she had had a normal mammogram 1 year before the current presentation and a normal breast examination during the year before the current presentation, findings that make breast cancer unlikely. Her weight loss and intermittent diarrhea were suggestive of a gynecologic cancer.

To establish a diagnosis in this patient, I requested a cytologic analysis of the pleural fluid and CT scans of the chest. I then had a discussion with the patient about navigating the time of uncertainty while awaiting the results of diagnostic testing.

NAVIGATING UNCERTAINTY

Sitting with our patients during a period of uncertainty is a large part of our work in continuity care. Finding out what our patients want to know, and how they want to hear it, is important. In an era of patient portals and immediate access to test results, staying on top of test interpretation is especially important. Results come quickly and can be confusing (for us and for patients). Research has shown that patients find immediate access to test results through patient portals to be acceptable and even preferable to waiting to discuss the results with a clinician. However, in many cases, the test results will not be easily interpretable, by the patient or the clinician, yet are available for viewing in the portal as soon as they are finalized.

In this patient's case, I told her that one of my main concerns was that she had cancer and that we would probably not know the diagnosis until the results of the cytologic evaluation of her pleural fluid were available. I indicated that I would call her once I had the results. I did, however, have a nagging worry that she would see a diagnosis of cancer reported before I had a

chance to review the results myself — a common cause of stress for those of us in primary care practice. Although there is growing interest in the development of patient-centered pathology reports that are written in clearer language for the lay reader, such reports are not yet widespread.^{4,5}

DR. LEIGH H. SIMMONS'S DIAGNOSIS

Exudative pleural effusion, most likely due to cancer.

INITIAL DIAGNOSTIC TESTING

Dr. Megan J. Fitzpatrick: The numerous unclassified cells detected in the pleural fluid were flagged for pathological review by the hematology laboratory. These cells were found to be plasma cells, including atypical forms with nucleoli and binucleation.

OUTPATIENT COURSE

Dr. Simmons: One of my challenges in this case was that, as the patient's test results became available, I became less certain about her diagnosis rather than more certain. The identification of the unclassified cells as plasma cells was reported before additional diagnostic testing had become available. When I received the result indicating the presence of plasma cells in the pleural fluid, I called a colleague who is an expert in myeloma. This colleague suspected that the source of the plasma cells was a plasmacytoma on one of the patient's ribs and recommended proceeding with serum protein electrophoresis and measurement of the β_2 -microglobulin level. I referred the patient to our myeloma center to undergo this testing. The patient and I then awaited the results of chest CT and the cytologic analysis of the pleural fluid.

ADDITIONAL DIAGNOSTIC TESTING

Dr. Willett: CT of the chest, performed after the intravenous administration of contrast material, revealed a moderate-to-large right pleural effusion with adjacent atelectasis. There was complete consolidation of the right middle lobe with mucus plugging of the bronchus of the right middle lobe (Fig. 2A and 2B).

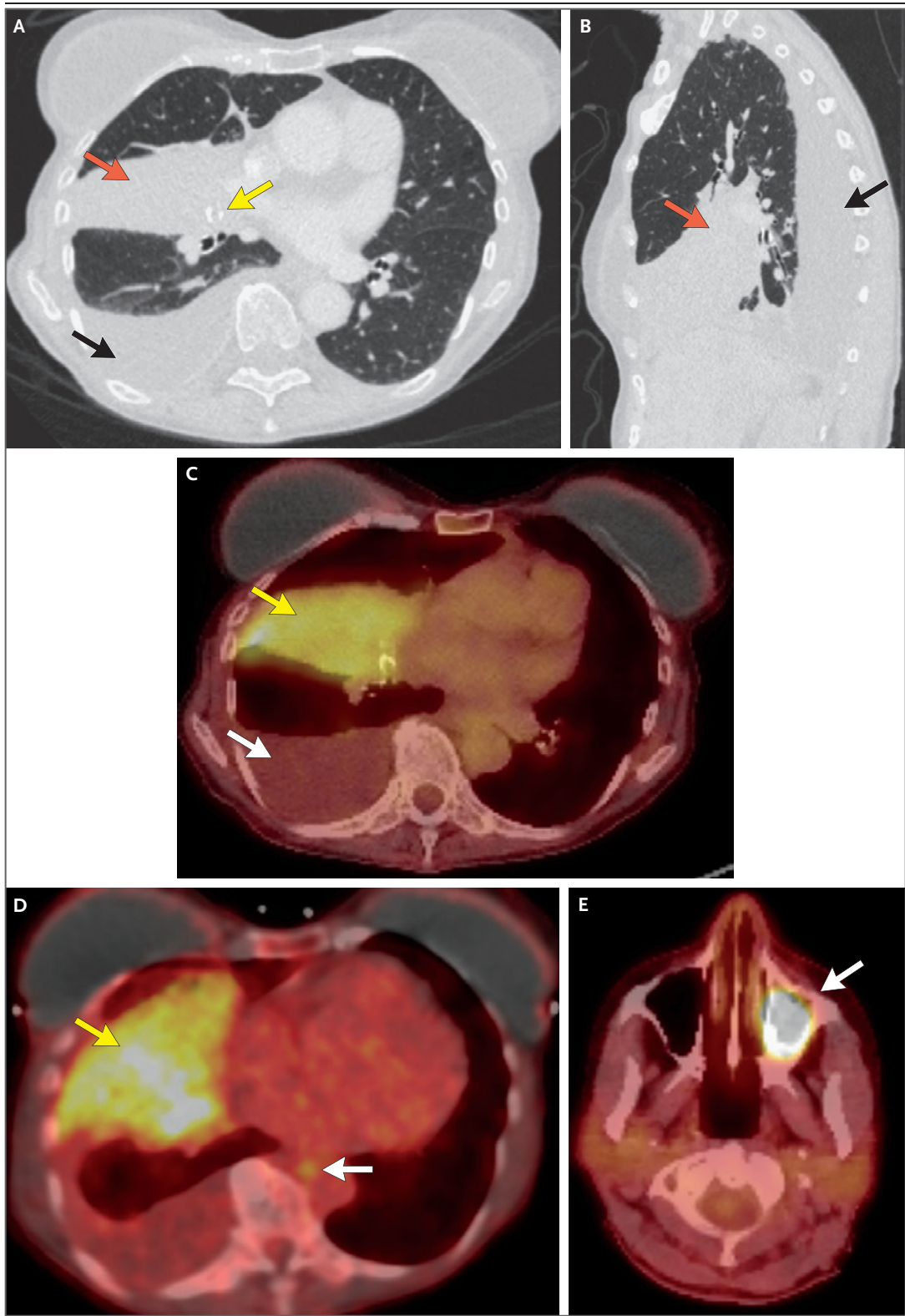


Figure 2 (facing page). CT and PET-CT Imaging Studies.

Axial and sagittal CT images of the thorax (Panels A and B, respectively) show a moderate-to-large right pleural effusion (black arrows) with adjacent atelectasis. There is complete consolidation of the right middle lobe (Panels A and B, red arrows) with mucus plugging of the bronchus of the right middle lobe (Panel A, yellow arrow). Images of the thorax obtained on positron-emission tomography with CT (PET-CT) (Panels C and D) show marked ^{18}F -fluorodeoxyglucose (FDG) uptake in the consolidation in the right middle lobe (yellow arrows) but no abnormal FDG uptake in the right pleural effusion (Panel C, white arrow). Moderate FDG uptake is present in a few thoracic lymph nodes (Panel D, white arrow). An image of the head obtained on PET-CT (Panel E) shows opacification with corresponding marked FDG uptake in the left maxillary sinus (arrow).

Dr. Fitzpatrick: Serum protein electrophoresis with immunofixation showed an IgM lambda M component (0.96 g per deciliter), and the serum free light-chain ratio (ratio of kappa to lambda light chains) was 0.10 (reference range, 0.30 to 1.70). The β_2 -microglobulin level was $3.43 \mu\text{g}$ per milliliter (reference range, 0.80 to 2.34).

Cytologic evaluation of the pleural fluid showed numerous mature-appearing plasma cells, including occasional atypical forms with Dutcher bodies (Fig. 3A and 3B). On immunohistochemical staining, the plasma cells showed marked IgM expression (Fig. 3C). Immunohistochemical staining showed a few small CD20-positive B cells, which were negative for CD5 and were admixed with more numerous CD3-positive small T cells. On flow cytometry of the pleural fluid, plasma cells maintained expression of CD45 and CD19 (Fig. 3E), were negative for CD56, and showed a marked excess of lambda immunoglobulin light chain (kappa:lambda ratio, 0.02) (Fig. 3D), findings consistent with a monocytic plasma-cell population. B cells also showed a mild excess of lambda light chain (kappa:lambda ratio, 0.6) (Fig. 3F), which suggested the possibility of a concurrent monocytic B-cell population. Polymerase-chain-reaction (PCR) testing for the MYD88 L265P mutation was negative.

The overall findings of a monocytic plasma-cell population with maintenance of CD19 and CD45 expression and lack of CD56 expression, in conjunction with the IgM expression on plas-

ma cells on immunohistochemical staining and the excess of lambda light chain on B cells, favored a diagnosis of low-grade B-cell lymphoma with marked plasmacytic differentiation rather than a diagnosis of IgM-positive plasma-cell myeloma.^{6,7}

Flow cytometry of the peripheral blood showed a minute lambda-restricted monocytic plasma-cell population (accounting for <1% of the total white-cell count). A small lambda-restricted monocytic B-cell population (accounting for 1% of the total white-cell count) was also noted.

A bone marrow biopsy was performed, and examination of a core biopsy specimen revealed slightly hypercellular marrow for the patient's age with maturing trilineage hematopoiesis (Fig. 4A). On immunohistochemical staining for CD138, plasma cells constituted less than 5% of the marrow cellularity. A 400-cell differential count of the bone marrow aspirate showed 2% plasma cells. Lymphocytes were not overtly increased, and no lymphoid aggregates were identified. On in situ hybridization, plasma cells showed an excess of lambda light chain, and on flow cytometry, a marked predominance of lambda light chain was present (kappa:lambda ratio, 0.2); these findings were consistent with a monocytic plasma-cell population (Fig. 4B, 4C, and 4D). B cells were not increased and showed a minimal excess of lambda light chain on flow cytometry (kappa:lambda ratio, 0.8), findings that are not diagnostic for the presence of a concurrent monocytic B-cell population (Fig. 4F).

PCR testing for immunoglobulin heavy-chain gene rearrangement was positive for a clonal B-cell population. A PCR test for the MYD88 L265P mutation was again negative, and a comprehensive next-generation sequencing panel detected no known pathogenic mutations.

Dr. Willett: Positron-emission tomography with CT (PET-CT) of the head, chest, abdomen, and pelvis was performed. Intense ^{18}F -fluorodeoxyglucose (FDG) uptake was present in the consolidation in the right middle lobe (Fig. 2C and 2D). No abnormal FDG uptake was noted in the right pleural effusion. Moderate FDG uptake was present in a few thoracic lymph nodes. A mass with corresponding marked FDG uptake was noted in the left maxillary sinus (Fig. 2E).

Dr. Fitzpatrick: A biopsy of the mass in the left maxillary sinus was performed. Examination of the biopsy specimen revealed an oncocyctic sinonasal papilloma with an atypical lymphoplasmacytic infiltrate (Fig. 5A and 5B). Lymphocytes consisted predominantly of small and mature CD20-positive B cells that occurred in clusters and were occasionally associated with CD21-positive meshworks of follicular dendritic cells (Fig. 5C and 5D). There was no B-cell coexpression of CD5, BCL6, or cyclin D1. On immunohistochemical staining, an increased number of mature plasma cells was present

with excess expression of IgM heavy chain and lambda immunoglobulin light chain (Fig. 5E and 5F).

In summary, the overall findings from examination of multiple specimens were most consistent with a diagnosis of low-grade B-cell lymphoma with plasmacytic differentiation. The absence of immunophenotypic aberrancy on plasma cells, cyclin D1 expression, bone marrow involvement, and other myeloma-defining events — in addition to the persistent excess of lambda light chain on B cells — argued against a diagnosis of IgM-positive plasma-cell myeloma.

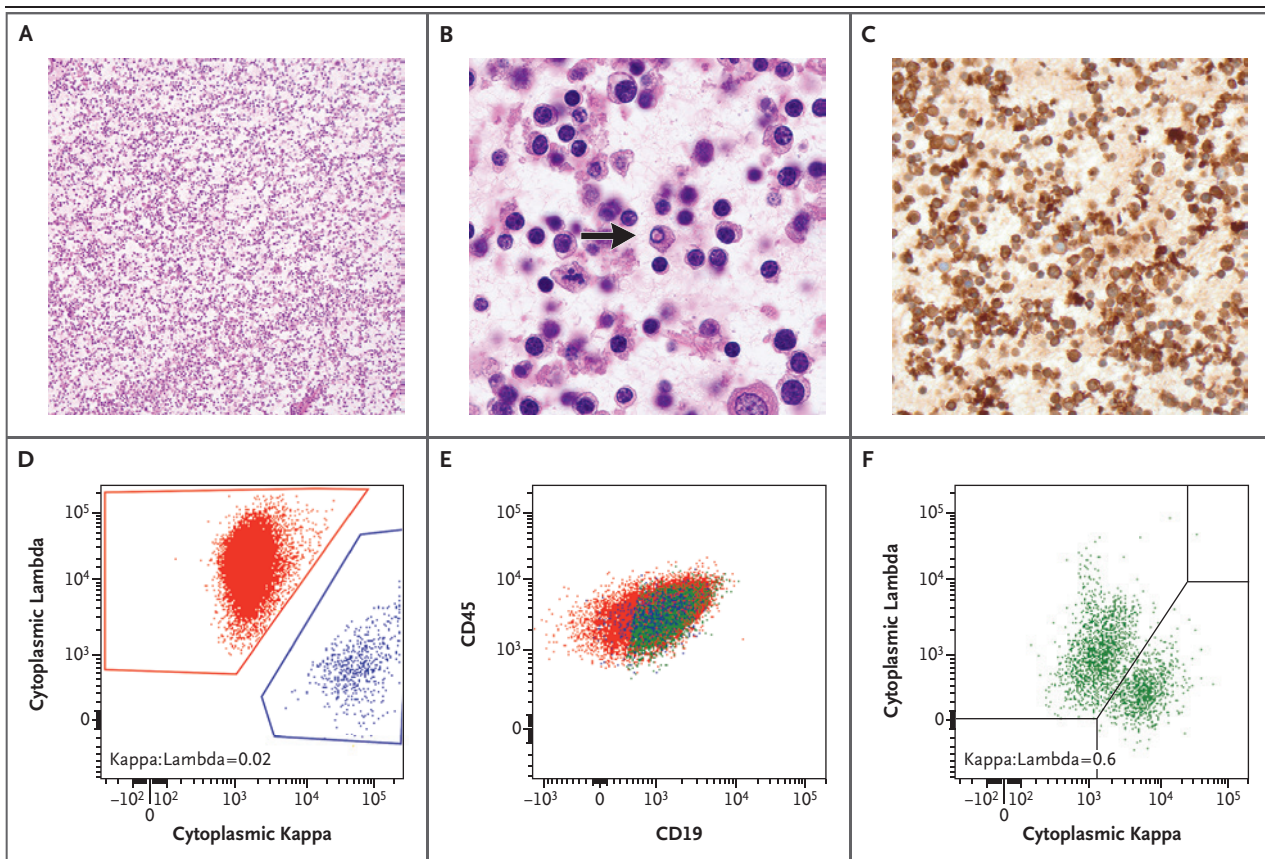


Figure 3. Pleural Fluid Specimen and Flow-Cytometric Studies.

A cell-block preparation of the pleural fluid was made. Hematoxylin and eosin staining (Panel A) shows high cellularity with a plasma-cell predominance. At higher magnification (Panel B), numerous mature-appearing plasma cells with clumped chromatin, eccentric round nuclei, and moderate amounts of cytoplasm with occasional Dutcher bodies are seen (arrow). On immunohistochemical staining (Panel C), plasma cells show expression of IgM heavy chain. Flow cytometry of the pleural fluid specimen was performed. Numerous CD38-positive plasma cells with a marked excess of lambda immunoglobulin light chain are present, findings consistent with a monotypic plasma-cell population in a polytypic background (Panel D; kappa-positive plasma cells shown in blue, lambda-positive plasma cells shown in red). Plasma cells show no overt loss of CD45 or CD19 in comparison to B cells (Panel E; B cells shown in green). B cells also show a mild excess of lambda immunoglobulin light chain, a finding that is suggestive of a concurrent monotypic B-cell population in a polytypic background (Panel F; B cells shown in green).

The absence of an *MYD88* mutation and of pathologically significant bone marrow involvement was not consistent with a diagnosis of lymphoplasmacytic lymphoma. In addition, the patient did not have lymphadenopathy and splenomegaly, which argued against the presence of a nodal lymphoma as well as primary splenic marginal-zone lymphoma. Although the primary site of disease was not sampled, the overall findings were best classified as extranodal marginal-zone lymphoma that had possibly originated in the lung, given the intense FDG uptake seen in the consolidation on imaging.

PATHOLOGICAL DIAGNOSIS

Extranodal marginal-zone lymphoma.

DISCUSSION OF MANAGEMENT

Dr. J. Erika Haydu: Marginal-zone lymphoma is the third most common indolent non-Hodgkin's lymphoma; subtypes include extranodal, splenic, nodal, and cutaneous.^{8,9} Extranodal marginal-zone lymphoma of mucosa-associated lymphoid tissue (also known as mucosa-associated lymphoid tissue [MALT] lymphoma) is the most prevalent

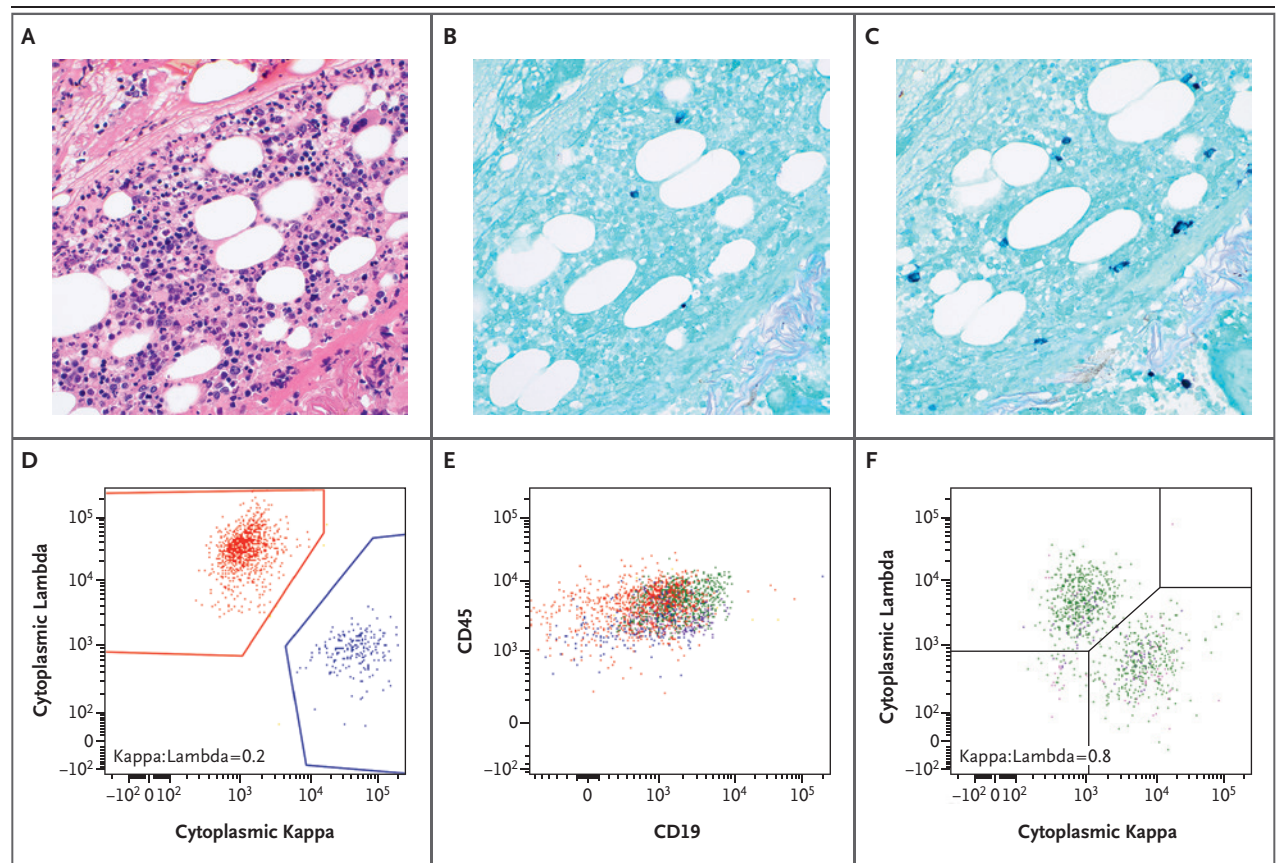


Figure 4. Biopsy Specimen of the Bone Marrow.

Hematoxylin and eosin staining of a core biopsy specimen (Panel A) shows slightly hypercellular marrow for the patient's age (approximately 40%) with maturing trilineage hematopoiesis and no overt increase in plasma cells or lymphocytes. Although plasma cells account for less than 5% of the marrow cellularity on immunohistochemical staining for CD138 (not shown), in situ hybridization for kappa (Panel B) and lambda (Panel C) immunoglobulin light chains shows an excess of lambda light chain on plasma cells, a finding consistent with a monocytic plasma-cell population in a polytypic background (Panel D; kappa-positive plasma cells shown in blue, lambda-positive plasma cells shown in red). Plasma cells do not show overt loss of CD19 or CD45 as compared with B cells (Panel E; B cells shown in green). B cells show a minimal excess of lambda light chain, a finding that is not diagnostic for the presence of a concurrent monocytic B-cell population involving the bone marrow (Panel F; B cells shown in green).

subtype; it most commonly affects the stomach and less commonly the ocular adnexa and lungs, but it can involve any extranodal location.¹⁰ Risk factors for extranodal marginal-zone lymphoma include autoimmune disease and infection — a classic subtype is *Helicobacter pylori*-associated gastric extranodal marginal-zone lymphoma, although associations with other organisms based on the specific site of disease can occur. Less common marginal-zone lymphoma subtypes include splenic, which is characterized by splenomegaly, and nodal; subtype overlap can occur. A family history of lymphoma, as this patient had, is associated with a modestly increased risk of marginal-zone lymphoma.^{11,12}

This patient was assessed as having stage 4 marginal-zone lymphoma, also known as advanced-stage marginal-zone lymphoma, on the basis of involvement of multiple extranodal sites (lung, sinus, and bone marrow) and nodal sites.¹³ A diagnosis of marginal-zone lymphoma generally confers an excellent prognosis, with more than half of affected patients still alive 10 years after diagnosis.¹⁴ Prognostic calculators such as the MALT Lymphoma Prognostic Index can help stratify patients according to their risk of death.¹⁵ Marginal-zone lymphoma that progresses within 24 months after initial treatment portends a poorer prognosis.¹⁶⁻¹⁸ Marginal-zone lymphoma can undergo histologic high-grade transformation

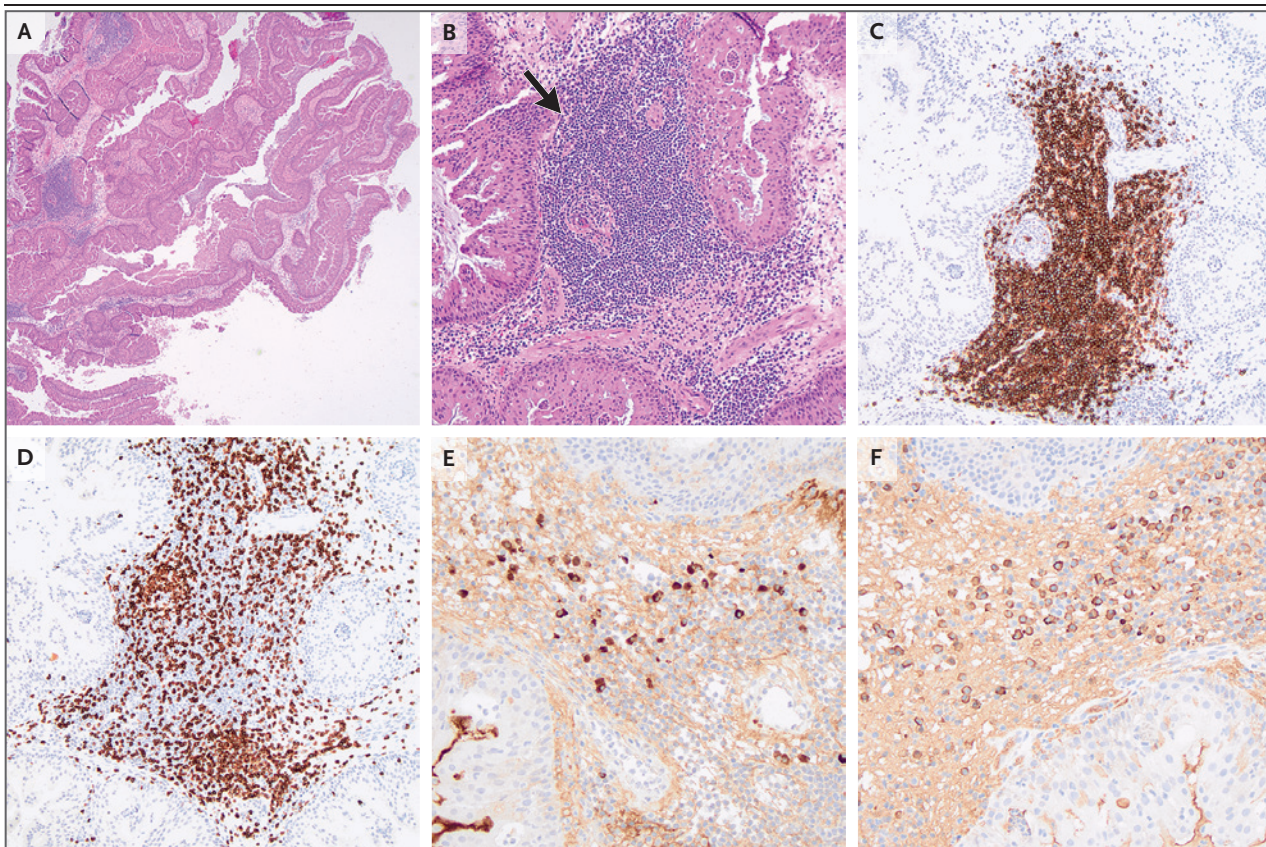


Figure 5. Biopsy Specimen of the Sinonasal Mass.

Hematoxylin and eosin staining of a biopsy specimen of the sinonasal mass (Panel A) shows an oncocytic sinonasal papilloma with an inverted growth pattern and atypical lymphoplasmacytic aggregates within the lamina propria. Higher magnification of the atypical lymphoplasmacytic infiltrate (Panel B) shows abundant small and mature lymphocytes with condensed chromatin, ovoid and irregular nuclei, and moderate amounts of cytoplasm. Mature-appearing plasma cells are admixed with small lymphocytes and also occur in clusters (Panel B, arrow). Immunohistochemical staining (Panels C through F) shows that the lymphoid aggregates are composed predominantly of small CD20-positive B cells (Panel C), which are more numerous than CD3-positive T cells (Panel D). Immunohistochemical staining for kappa (Panel E) and lambda (Panel F) immunoglobulin light chains shows an excess of lambda light chain on plasma cells.

to diffuse large B-cell lymphoma, with a cumulative incidence of 2.95 to 8% at 10 years after diagnosis.^{19,20}

Initial management of marginal-zone lymphoma depends on several factors.²¹ Treatment of *H. pylori* infection can cure gastric extranodal marginal-zone lymphoma in a majority of cases,²² but antiinfective agents are typically not otherwise used as primary treatment. Radiation therapy can be a curative approach for limited-stage marginal-zone lymphoma involving any site and is therefore prioritized when possible.²³⁻²⁵ Advanced-stage marginal-zone lymphoma, which this patient had, is technically incurable, although it remains highly treatable. Active surveillance is used if the disease is otherwise asymptomatic and nonbulky (<7 cm) because there is no overall survival benefit to early initiation of treatment in such cases. Symptoms are dependent on the pattern and extent of disease and can include those related to involved organs or lymph nodes (or both), pain, cytopenias, systemic symptoms (fevers, night sweats, and weight loss), or a combination of these.²⁶⁻²⁸ Advanced-stage marginal-zone lymphoma with a low tumor burden that warrants therapy can be treated effectively with a time-limited course of the anti-CD20 monoclonal antibody rituximab.^{29,30}

In cases of disease with a higher tumor burden that necessitates treatment, a 4-to-6-month course of rituximab in combination with bendamustine results in high overall response rates (>90%) that are durable, with more than two thirds of patients in remission 4 years after initial treatment.³¹⁻³³ Treatment with bendamustine can result in prolonged cytopenias and an increased risk of infection, so patients should be counseled accordingly, and prophylaxis against *Pneumocystis jirovecii*, herpes simplex virus, and varicella-zoster virus infections is recommended. Maintenance treatment with rituximab after chemoimmunotherapy can delay relapse but does not have a survival benefit; thus, the potential benefits of such treatment must be weighed against possible toxic effects, including the risk of infection.³⁴

For this patient, given the extent of disease as indicated by the presence of a large symptomatic pleural effusion, we discussed initiation of standard-care chemoimmunotherapy (bendamustine and rituximab) or participation in a clinical

trial that did not involve chemotherapy (instead, involving treatment with the CD20–CD3 bispecific antibody glofitamab and the CD20 monoclonal antibody obinutuzumab; ClinicalTrials.gov number, NCT05783596). She opted to participate in the clinical trial. The treatment course was complicated by mild cytokine release syndrome after an initial increase in the glofitamab dose. The patient initially underwent frequent drainage of the pleural effusion, but the effusion, as well as dyspnea and diarrhea, resolved within the first 2 months of treatment. A restaging study with PET after 2 months of treatment with glofitamab and obinutuzumab revealed a complete metabolic response, and the patient continues to receive the protocol-based therapy.

PATIENT PERSPECTIVE

The Patient: After experiencing coughing, I underwent a number of tests. I didn't realize how long it could take for a final diagnosis to be made — a particular concern when it initially appeared that I might have an aggressive form of cancer. Nevertheless, I found the tests and analyses, which were carried out over the next few months, along with the explanations provided to my family and me, to be both logical and informative. I also appreciated the efforts of the oncologists and other medical professionals with whom I met to educate and inform us, particularly given the complexities involved in making a final diagnosis.

Because of my illness, I became more knowledgeable about cancer in general and several forms of lymphoma in particular, and I took advantage of a number of useful online resources, especially those provided by the National Cancer Institute (<https://www.cancer.gov/publications/patient-education>). In addition, I was particularly impressed by the ability of the oncologist I met with at my very first visit to outline the treatment that would most likely be prescribed, including the medications, the length of treatment, and the anticipated outcome. The oncologist's empathetic approach, as well as that of others I encountered during initial and subsequent visits, made such a difference, particularly before receiving a final diagnosis. As the number of tests continued to grow, I found the trend graphs that showed my test results over time to be especially helpful and

generally quite comforting, since they allowed me to see the big picture.

After my diagnosis, I was able to enter a clinical research trial with access to medications that would otherwise be unavailable to me. I am very grateful for this opportunity, but I didn't fully appreciate the extent of the adverse effects that could occur. I would encourage medical providers to ensure that the range of such effects, to the extent that they are anticipated, are clearly understood by the patient, particularly if that patient has other responsibilities or must make plans for his or her care.

FINAL DIAGNOSIS

Extranodal marginal-zone lymphoma.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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